ENGINEERING SCIENCE

Revised Sept 2020

Pupil booklet

Mechanisms and Structures:

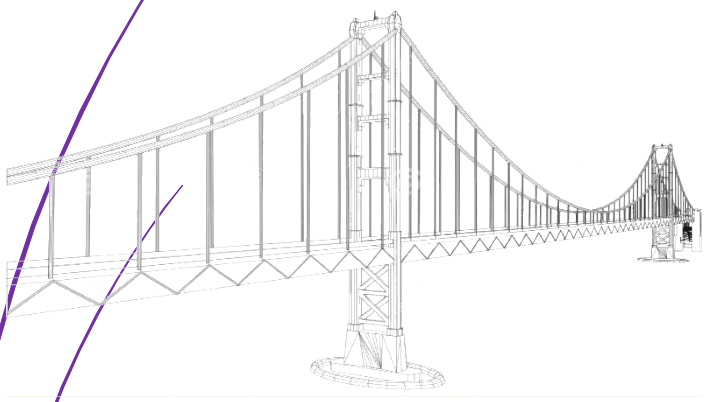
Levers, Structures and Vectors

Topic 1: Levers

Topic 2: Structures and Moments

Topic 3: Vectors

Topic 4: Materials





Name: _____

Class: _____

Teacher: _____

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Outcomes and assessment standards

Outcome 1

The learner will:

1 Investigate a range of mechanical and pneumatic systems by:

1.1 Describing or producing diagrams of a range of structures.

1.2 Describing or producing diagrams of a range of pneumatic systems

1.3 Describing or producing diagrams of a range of mechanical drive systems.

1.4 Carrying out calculations involving energy, work, power and efficiency using given formulae

Outcome 2

The learner will:

2 Develop mechanical or pneumatic solutions to solve problems by:

2.1 Identifying key aspects of a problem.

2.2 Applying knowledge and understanding of structures, pneumatics and/or mechanical drive systems.

2.3 Designing, and simulating or building, mechanical or pneumatic systems

2.4 Testing and evaluating solutions against a specification

Use traffic lighting to record your progress. This will also help with revision as you will be able to see what areas you need to work on. Your teacher will sign off this page after speaking to you about your self-evaluation:

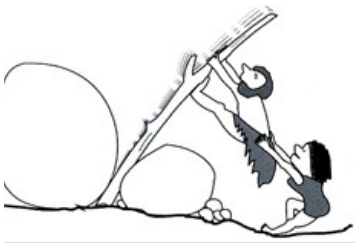
Red:	I need to work on this topic further to understand it.
Amber:	I need a bit of help to fully understand this topic.
Green:	I am confident in my understanding of this topic.

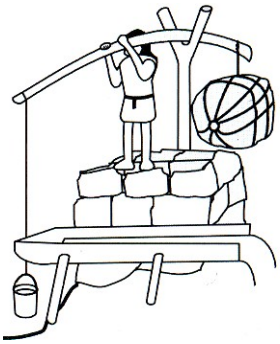
Success Criteria

- ☐ I can produce diagrams to show the change in motion.
- ☐ I can convert motion using different devices such as levers.
- ☐ I understand the terms; equilibrium, reaction forces, friction, torque and efficiency.
- ☐ I can produce diagrams of a range of structures.
- ☐ I can use the moments calculation to solve problems.
- ☐ I can understand and can describe the various parts of a structure.
- ☐ I am able to apply knowledge and understanding of key engineering facts and ideas.
- ☐ I can apply skill in analysis, design, construction and evaluation to a range of straightforward engineering concepts.
- ☐ I am able to apply critical thinking when evaluating the quality and effectiveness of my own and other peoples products and systems.

Levers

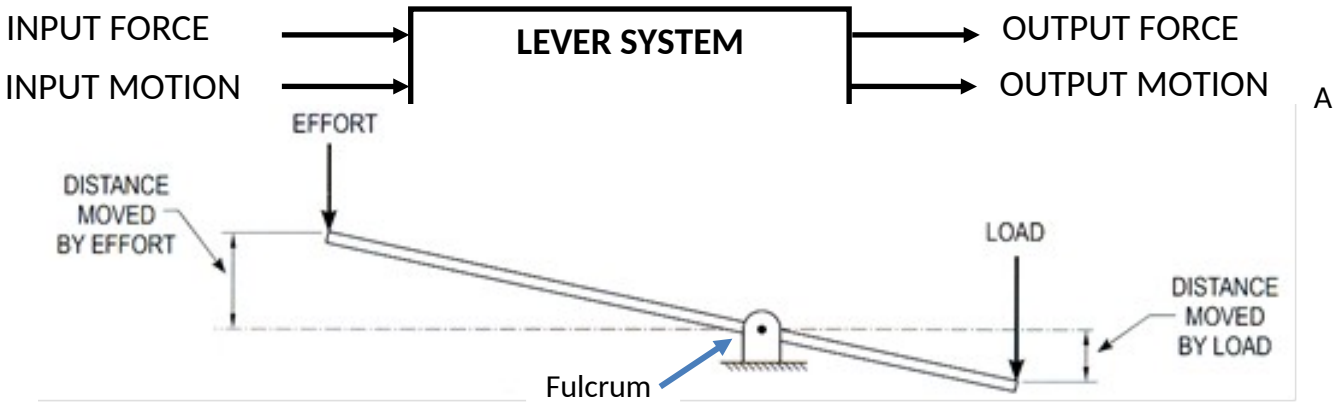
The picture shows an early lever. The large boulder is too heavy to move by pushing it. By using a small boulder as a pivot point and a branch as a lever, it is possible to increase the force applied to the large rock. The further from the pivot the effort is applied, the easier it is to move the large rock or load.





When a weight is attached to one side of a lever to assist the user, it is known as a '**counterbalance**'

A universal system diagram of a lever is shown below. A lever system changes an input force and an input motion into an output force and an output motion.



A

- The output force is called the load
- The output motion is the distance moved by the load

The lever is a mechanism which makes things easier as it just a multiplier of the force (effort) and is normally used to get a large output force from a small input force. However, it can also be used as a 'distance' multiplier, giving a large output movement for a small input motion; but it cannot do both at the same time.

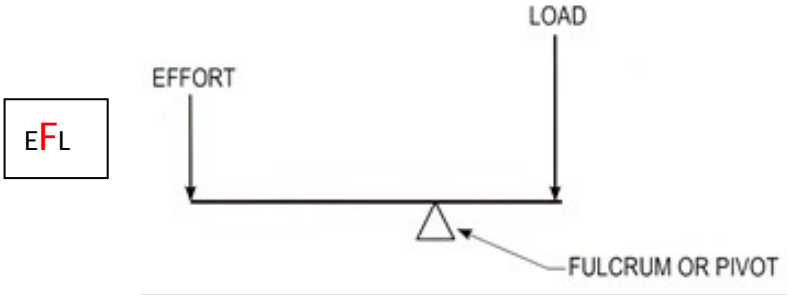
- The point that a leaver pivots about is called a fulcrum.
- The input force is called the effort.
- The input motion is the distance moved by the effort force.
- The output force is called the load.
- The output motion is the distance moved by the load.

Classes of levers

Levers can be divided into 3 distinct types of classes. The class is determined by the position of the load, effort and fulcrum. The load is the scientific name for the weight. The effort is quite simply the amount of effort used to push down on the rod in order to move the weight. The following classification diagrams have been simplified and are called free body diagrams. The fulcrum is replaced by a triangle to illustrate a knife edge of balance.

Class 1

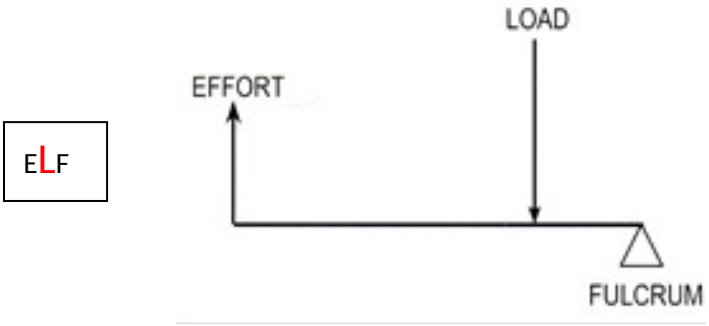
In this type of lever, the effort is on one side of the fulcrum and the load is on the opposite. Class 1 levers are the easiest to understand: the longer the crowbar the easier it is to prise open the lid.

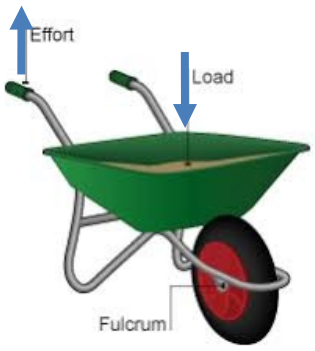




Class 2

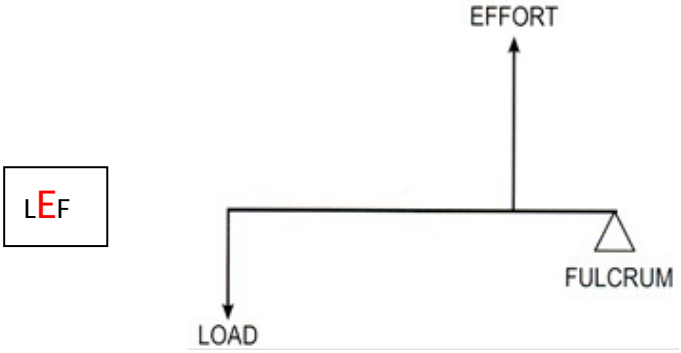
In this type of lever, the fulcrum is at one end of the lever and the load and the effort are spaced out at the other end of the bar. The load must be closer to the fulcrum than the effort. An example of a class 2 lever is a wheelbarrow, where the wheel is the fulcrum, the load is in the container area, and the effort is applied at the handles.





Class 3

Class 3 levers are similar to class 2 levers, except that now the effort is closer to the fulcrum than the load. This means more effort has to be applied to move the load. This type of lever is used when mechanisms require a large output movement for a small input movement.





This can be remembered by using the acronym **F.L.E.** The letter stands for whatever part is in the middle.

Fulcrum = class 1 **Load = class 2** **Effort = class 3**

1. Identify the following lever examples.

i. Draw a free body diagram with arrows showing the fulcrum, effort and load.
ii. Name the class of lever.

A)



B)



C)

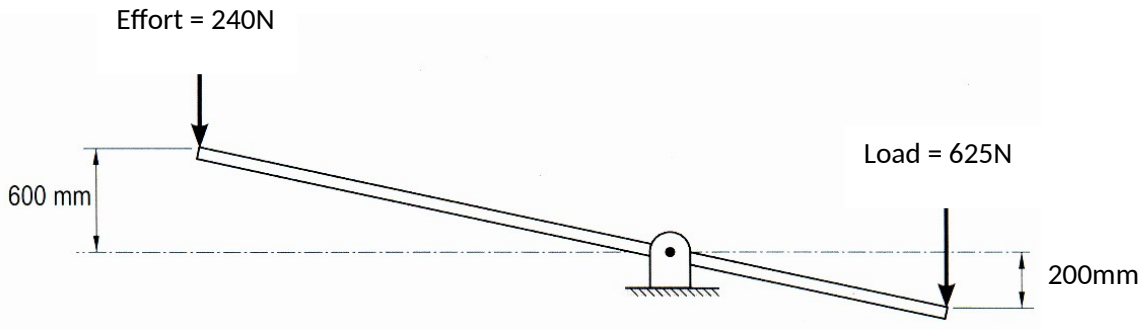


D)



2. Use the website 'Technology student.com' and locate the heading named Forces / Moments. That aspect of the website is concerned with the different types of forces that can be applied to any structure. Moments of a force and equilibrium are also discussed. Your teacher may provide you with some worksheets to complete.

Force Multiplier Ratio



Above shows a lever system designed to move heavy objects to a higher position. The load that is being lifted is about 3 times more than the effort being applied. The load divided by the effort gives a ratio, and this ratio is a **force multiplier**. In other words, how much more load can be lifted compared to the effort.

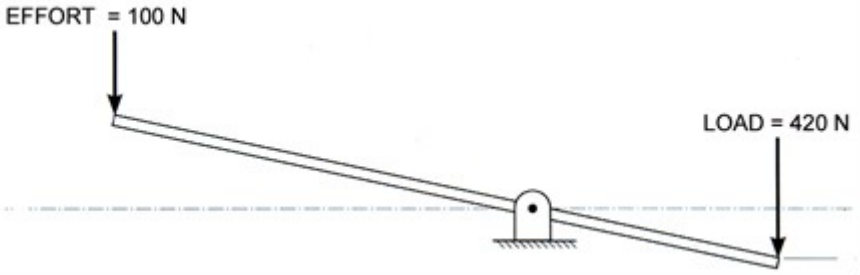
Worked example:

$$\text{Force multiplier ratio} = \frac{\text{Load}}{\text{Effort}} = \frac{625\ \text{N}}{240\ \text{N}} = 2.60 \quad \text{or } 2.6:1$$

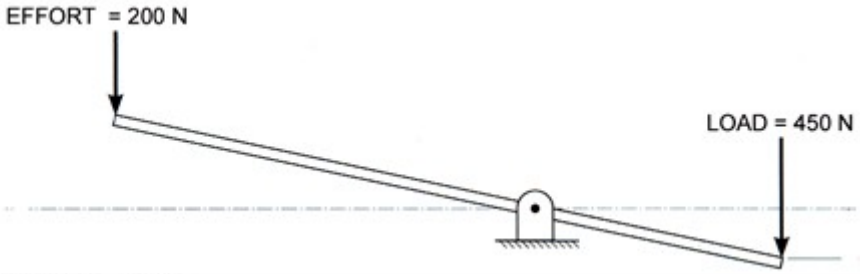
The lever therefore has a force-multiplier ratio of **2.6:1** (a ratio has **no** units of value)

3. Copy the following diagrams as free body diagrams into your jotter and calculate the **force multiplier ratio**. Replace the fulcrum with a triangle. Use a different colour of pen/pencil to identify the equilibrium position.

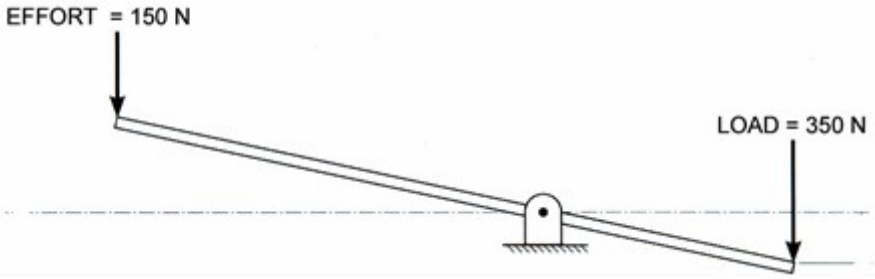
a)



b)



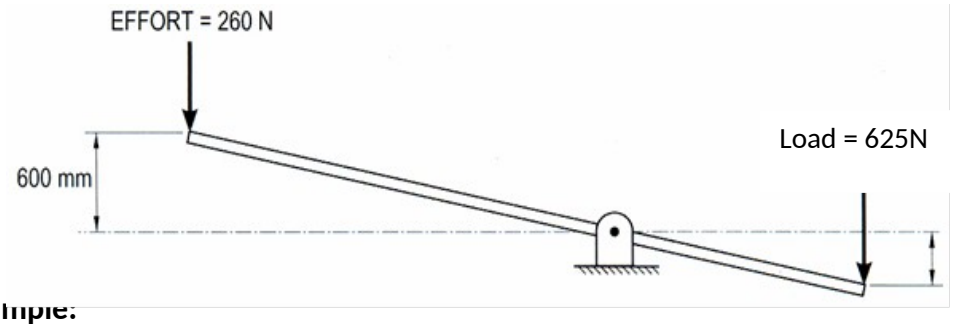
c)



Movement-Multiplier Ratio

In the diagram below, the force multiplier ratio appears to give the user something for nothing. The user is only applying about a third of the force to move the load. However, it can be seen from the diagram that the effort side of the lever has to move much further than the load side. The ratio of the **distance** moved by the effort, divided by the **distance** moved by the load, is known as the Movement-Multiplier Ratio.

Effort = 240N



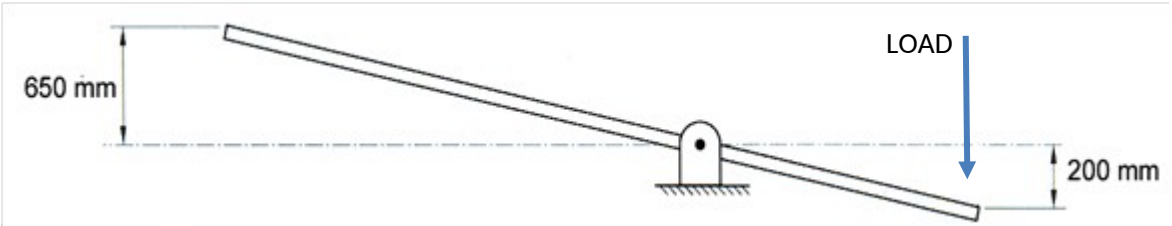
Worked example:

Movement-Multiplier ratio = **Distance** moved by the effort = 600
Distance moved by the load 200
= 3 or 3:1

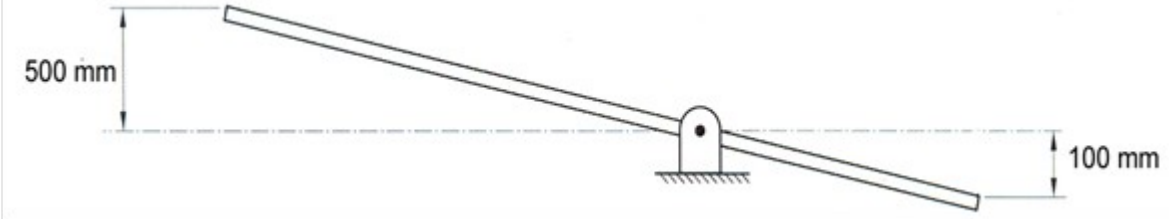
The lever diagram above therefore has a distance-multiplier ratio of **3** (again a ratio has **no units of value**).

4 Copy each diagram into your jotter and calculate the **Movement Multiplier Ratio**. Replace the fulcrum with a triangle. Use a different colour of pen/pencil to identify the equilibrium position.

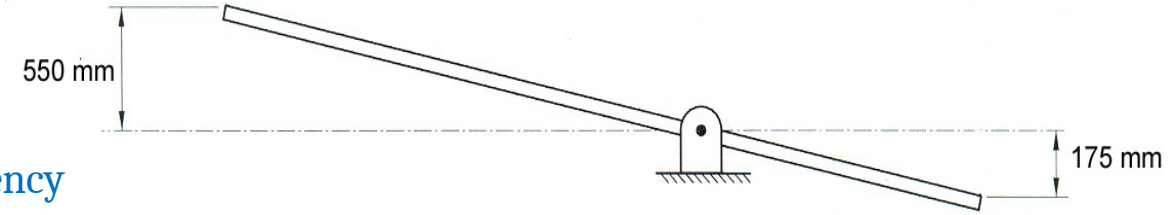
a)



b)



c)



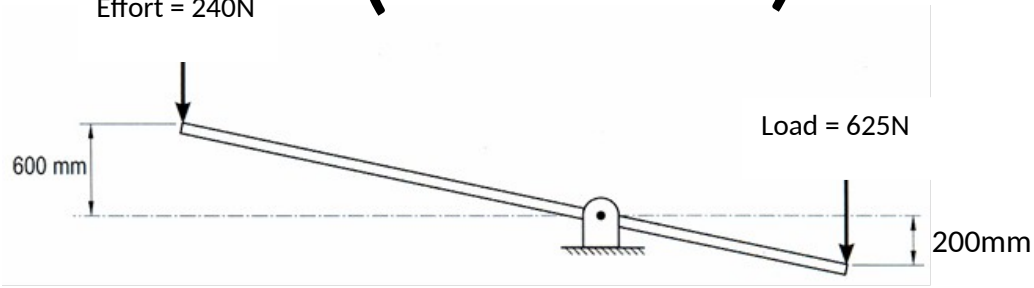
Efficiency

Owing to the effects of friction and inertia associated with the movement of any object, some of the input energy to a machine is converted into heat and losses occur, as learnt in unit 4 back in S3. Since losses occur, the energy output of a machine is less than the energy input; thus the mechanical efficiency of any machine cannot reach 100%.

Worked efficiency example:

$$\text{Efficiency } (\eta) = \left(\frac{\text{Force Ratio}}{\text{Movement Multiplier ratio}} \right) \text{X } 100$$

Effort = 240N



From our previous calculations with this diagram we know that:

Force multiplier ratio = 2.6

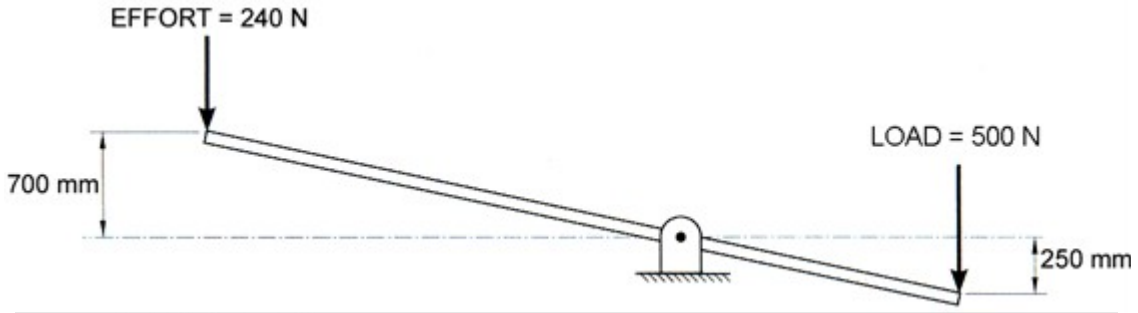
Movement multiplier ratio = 3

SO... $\eta = \left(\frac{2.6}{3} \right) \text{x } 100 = 0.866 \text{ x } 100 = \underline{\mathbf{86.6\%}}$

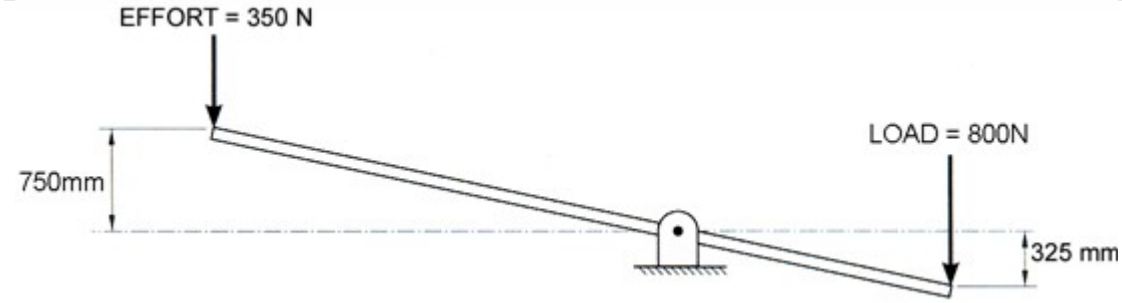
This system has an efficiency of 87%. No system can be 100% as there are always losses. The losses in a lever system consist of energy lost to friction at the fulcrum of the lever and the energy lost in strain as the lever bends slightly. In some cases, a small amount of energy will also be lost in the form of sound and heat.

5. Copy each diagram as a free body diagram into your jotter and calculate the **efficiency** (η). Replace the fulcrum with a triangle. Use a different colour of pen/pencil to identify the equilibrium position.

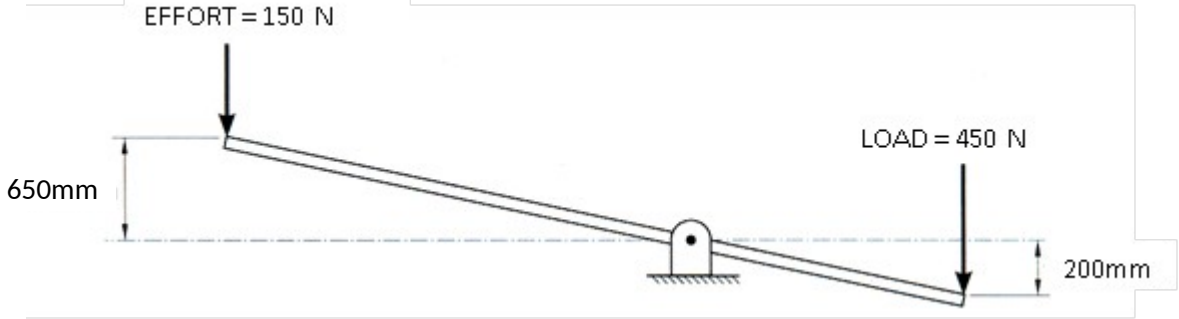
a)



b)



c)



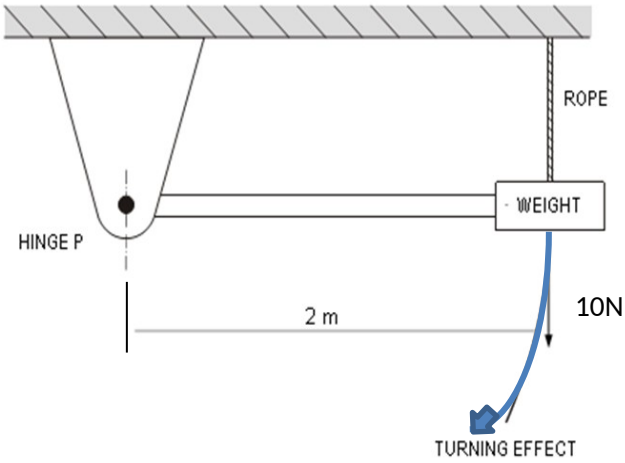
Moments

The image below shows a 10N stationary weight held in position by a rope. It is attached at the end of a metal rod and the rod is free to rotate around a hinge (pivot) P. If the rope is cut, the force on the weight causes the rod to swing or turn around the pivot.

The 'turning effect' that can occur on a structure to due force is called a **moment**. A moment is measured in Newton metres (Nm). The moment is calculated by the force multiplied by the distance turning the lever clockwise is equal to the force multiplied by the distance turning the lever anticlockwise.

The value of moment in this image is 20 Nm (10N x 2m).

As long as the rope is not cut, the weight and rod are held in balance by the force in the tie rope. When any system is in a steady state it is said to be in *equilibrium*.



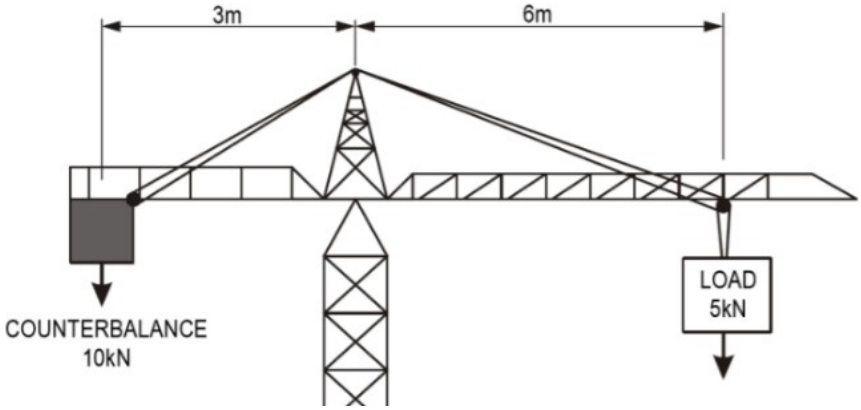
The principle of moments

The Greek letter ∑ (sigma) stands for 'the sum of' and can be used as a shorthand way of writing the principle of moments. It states that the 'sum of the Moments' (M) must equal zero **(ΣM = 0)** or the 'sum of the clockwise moments' must equal the 'sum of the anticlockwise moments' for a system to be in 'Equilibrium' (balanced and stationary).

Sum of ClockWise Moments = Sum of Anti ClockWise Moments

(∑CWM = ∑ACWM)

Large and complex mechanical systems, such as cranes, use moments to balance either side of the arm, otherwise the crane would topple over. When an item is perfectly balanced, it is called equilibrium. It is very important in the design of some structures to be in equilibrium or it will fail. The crane is in equilibrium as shown below:

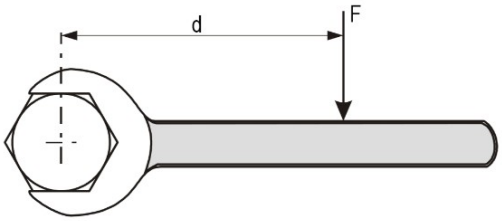


∑ACWM = ∑CWM

10KN × 3m = 5KN x 6m

30 KNm = 30 KNm

Using a lever, or the rule of moments, you can apply a small force with a short distance from the pivot to produce a turning force. This effect is effective when used to tighten or loosen a bolt with a spanner.



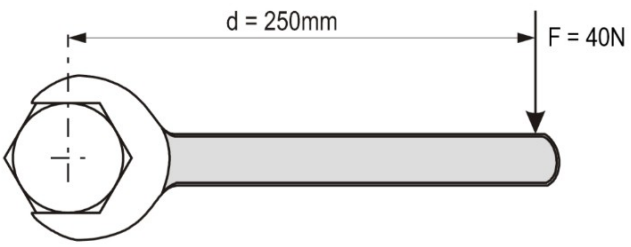
The effect of producing a large output force from a small input force is called a moment.

A larger moment is produced by moving the force further away from the pivot.

The moment of a force is dependent on the size of the force and the distance at which it is applied from the pivot point.

Worked example:

The spanner shown in the diagram is 250mm long. If a force of 40N is exerted at its end, find the moment of the force.

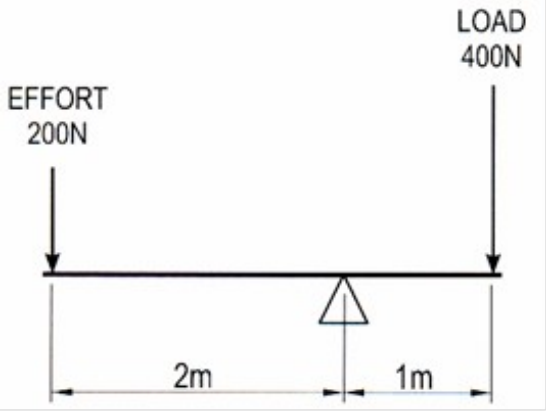


M = F × d

M = 40 × 0.250

M = 10 Nm

The lever system shown opposite is a lever that is in a state of equilibrium (balanced). The input effort is turning the lever Anti-Clockwise (AC); the load is turning the lever Clockwise (C). The forces on each end of the lever are exerting a moment: one clockwise, the other anticlockwise. If the beam (lever) is in equilibrium, **both** of these moments **must be equal**.



Worked example:

Using the lever system above, use the 'principle of moments' to show that the lever is in equilibrium.

For equilibrium to occur:

$$\sum CWM = \sum ACWM$$

$$F_1 \times d_1 = F_2 \times d_2$$

The load is exerting a ClockWise Moment; that is, it is tending to make the lever turn clockwise (CWM).

$\sum CWM = 400\ N \times 1\ m = 400\ Nm$

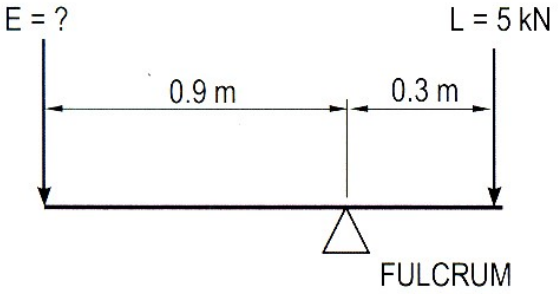
The effort is exerting an Anti ClockWise Moment (ACWM).

$\Sigma ACWM = 200\ N \times 2\ m = 400\ Nm$

Therefore: $\Sigma CWM = \Sigma ACWM$ (Therefore the lever is in a state of equilibrium.)

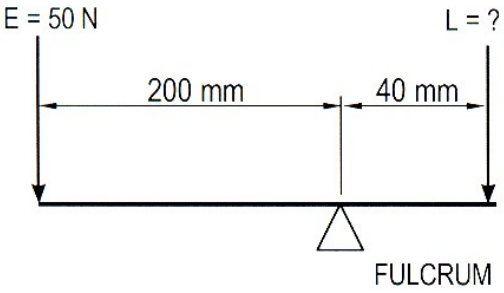
6. Copy each free body diagram into your jotter and calculate **the missing force or distance.**

A)



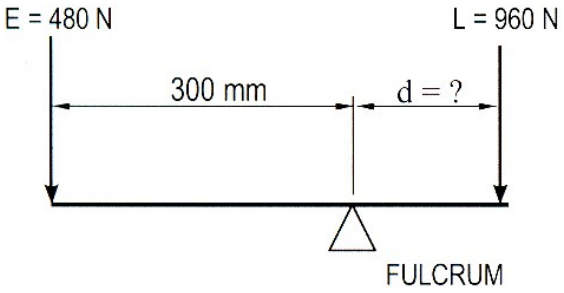
$\sum CWM = \sum ACWM$

B)



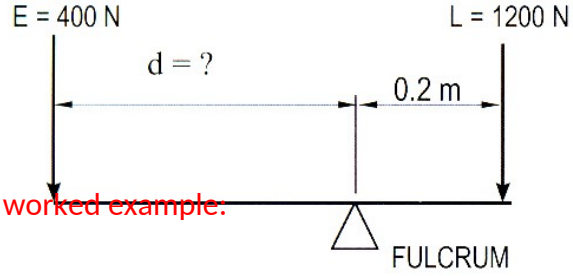
$\sum CWM = \sum ACWM$

C)



$\sum CWM = \sum ACWM$

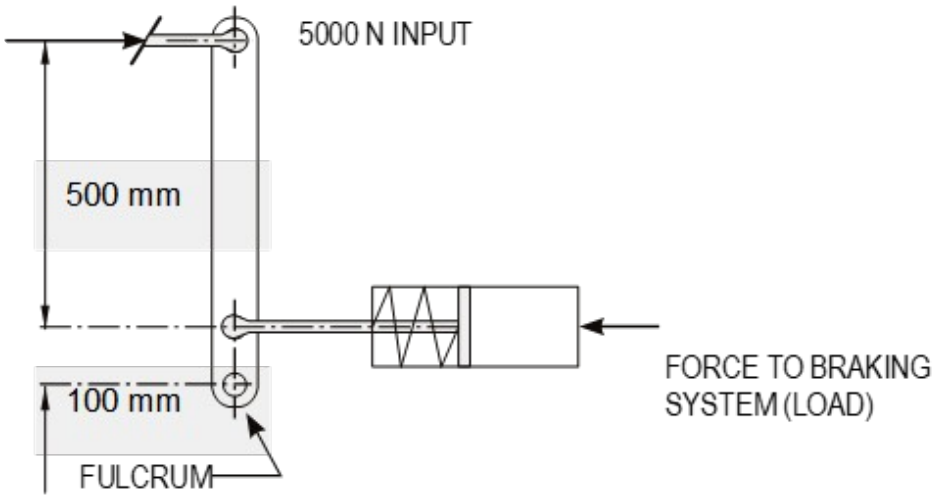
D)



$\sum CWM = \sum ACWM$

Teacher worked example:

A car footbrake uses a lever action to amplify the force transmitted by the driver to the braking system when the driver's foot presses the foot-pedal. If the driver's foot can exert a force of 5000 N, what force will be transmitted to the braking system?



This is a class 2 lever. Take moments about the fulcrum to find the force on the braking system. Notice the distance from the fulcrum to the input is 600 mm.

The input tends to make the lever turn clockwise; the braking system is opposing the input and so acts to turn the lever anticlockwise.

Teacher worked example:

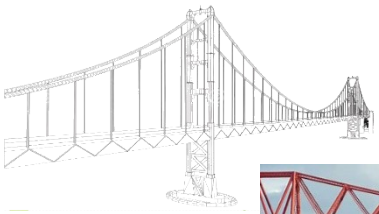
The principle of moments states that

$$\sum CWM = \sum ACWM$$

$$F_1 \times d_1 = F_2 \times d_2$$

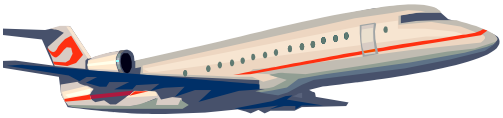
What is a structure?

Have a look at the pictures of the things below. Each one is a structure.









The five things all have something in common. These structures can all carry weight.

- A bridge carries the weight of everything travelling over it.
- The walls of your house hold the roof up over your head.
- Coke cans are stacked on top of each other when they are sold in supermarkets and also withhold the pressure of a carbonated juice.
- Egg boxes are also stacked on top of each other when they are in supermarkets.
- A passenger plane carries people long distances through the air.

What else can structures do?

Each of the five structures we looked at can carry a weight. But what else do they do?
There are FOUR jobs that structures can do for us:

- SUPPORT
- SPAN
- CONTAIN
- PROTECT.

7. Look at four of the structures again. Copy and complete the sentences below, using the four words stated above.



The walls of a house hold up the roof. They _____ the roof.



The bridge carries transport across a river or a valley. It _____the gap.



The can is filled with carbonated juice. It _____the liquid.



The egg box holds a number of eggs which break easily. It _____ the eggs from damage.

How are man-made structures built?

There is a huge number of man-made structures. They come in many different shapes and sizes.

8. What is the SMALLEST man-made thing you can think of?
What is the LARGEST man-made thing you can think of?

Identifying structures

The sizes of man-made structures vary a lot, but they come in three main forms;

- SOLID.
- FRAME.
- SHELL.

9. Look at the three structures below. For each of them, write down structure is and whether you think it is a solid structure, a frame or a shell.







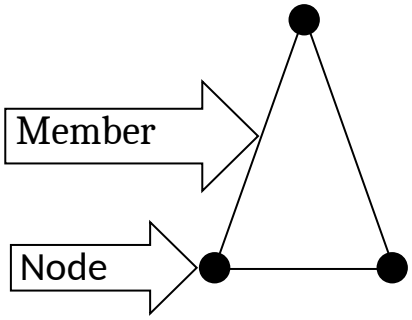
10. Copy the table into your jotter and identify their purpose. The first one is completed for you.

Structure	Man-made	Natural	Support	Span	Contain	Protect	Solid	Frame	Shell
Coke can	✓		✓		✓				✓
Feather									
Bookcase									
Water Bottle									
Eifel Tower									
Car									
Construction Crane									
Skeleton									

Frames

Frames are often very good ways of making structures. One of the main advantages of building with frames is that strong but light structures can be made. Frames are designed to carry a load.

A load will change the shape of a structure. The load could be a weight or a push or a pull. Usually they must also be made RIGID. This means they must not move or change shape very much when they are carrying the load. The change might be so small that you cannot see it happen.



The effect on a frame shape

In frame structures, the sides of the structures are called members.

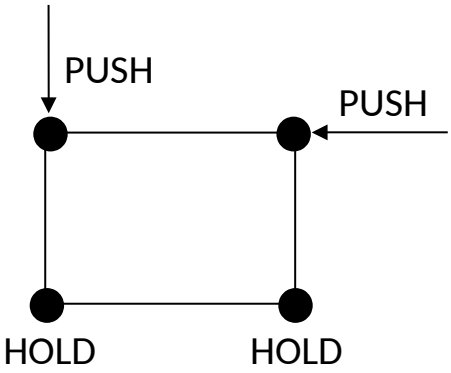
The points where members meet are called nodes.

11.

a) Make the square shape shown on the right. Lie the square flat on the bench. Hold down two of the corners. Gently push one of the corners not being held.

Does the frame move or change shape?

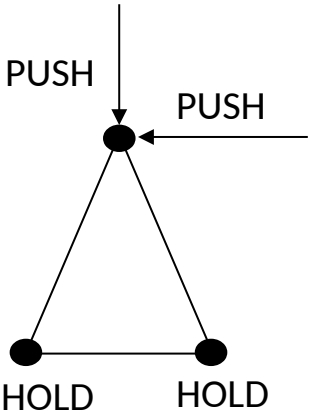
Copy the diagram into your jotter and record the results in full sentences.



b) Use the kits available to make a triangle. Lie the triangle flat on the bench. Hold down two of the corners. GENTLY push the third corner towards the other two. Then push it to the left.

Does the frame move or change shape?

Copy the diagram into your jotter and record the results in full sentences.

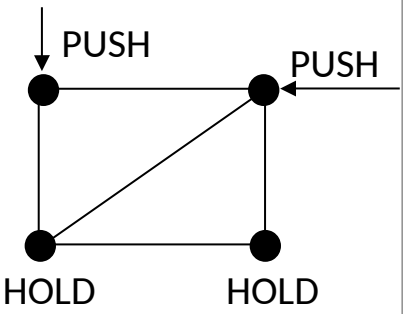


(c) Modify the shape, so it has a diagonal member. Lie the rectangle flat on the bench. Hold down two of the corners. GENTLY push one of the corners not being held. Try pushing in both the directions shown.

Does the frame move or change shape as much as without the diagonal piece?

Why do you think the square frame becomes more rigid when the diagonal member is added?

Copy the diagram into your jotter and record the results in full sentences.

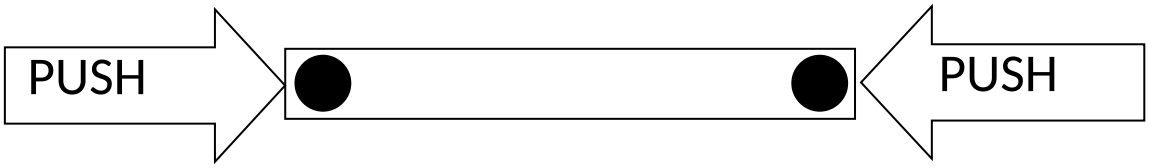


Triangulation

This is a technique used in frame structures to provide strength and support. When a force is applied to a triangular frame, two of its members stretch the third one, making it tense. This in turn pulls the other two members towards it, making the structure rigid and spreading the force between all three members of the triangle.

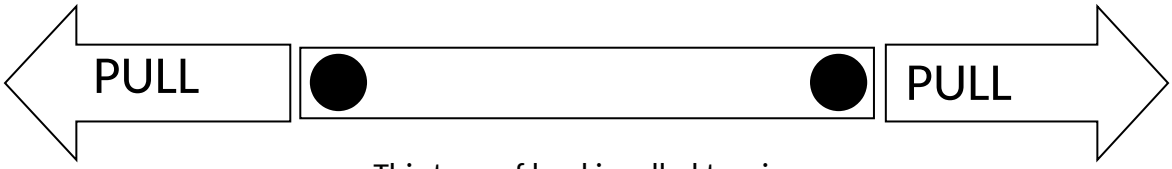
What happens to members when a frame is loaded?

1. If you push the ends of a member together it gets shorter and the middle gets fatter. Pushing is a type of load. If the member is being pushed, it is being compressed and it is called a strut.



This type of load is called compression

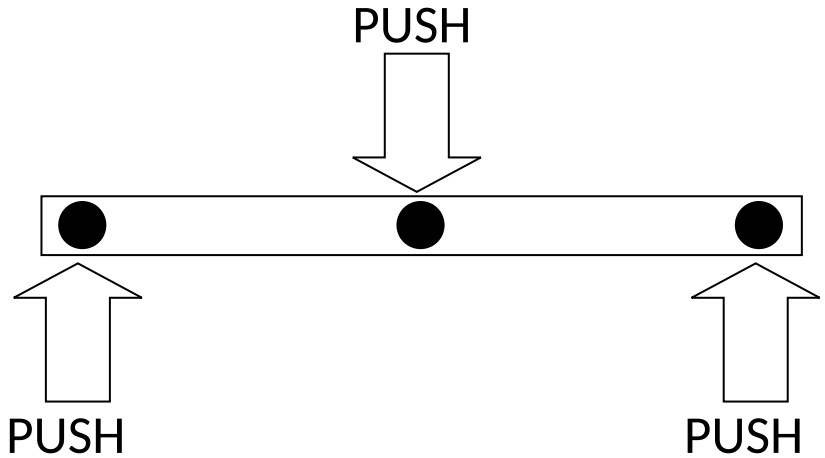
2. If you pull the ends of a member apart it gets longer and the middle gets thinner. Pulling is a type of load. If the member is being pulled, it is being STRETCHED and it is called a TIE.



This type of load is called tension

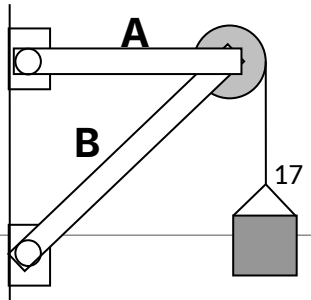
The load that a member can take before it breaks is called its strength.

If you bend a member the top face gets shorter and the bottom face gets longer.



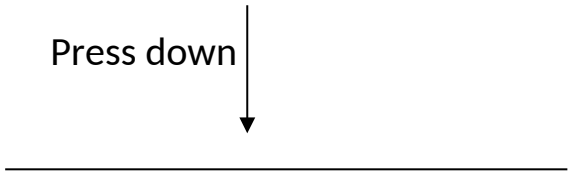
This type of force is called bending

12. The hoist is holding up a box. The hoist is made from two members fixed to a wall.

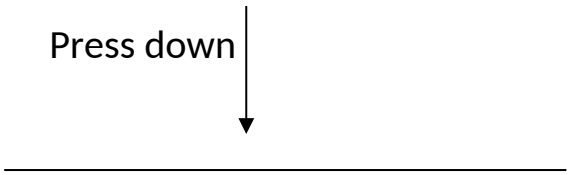


a) Which member is in compression?
b) Which member is in tension?
c) Which member could be replaced by a steel rope?

13. a) You will need 2 pencils and a ruler. Put a ruler on top of the 2 pencils and push down in the middle.

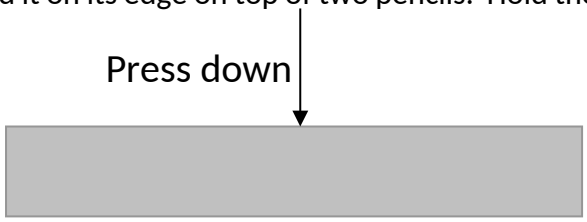


b) Now take the ruler and lay it on top of three pieces of dowel. Press down on the middle of the ruler.



ADDING A SUPPORT TO A BEAM WILL MAKE IT STIFFER

c) Finally, take the ruler and hold it on its edge on top of two pencils. Hold the ruler in this position and push down in the middle.



A THICK BEAM IS STIFFER THAN A THIN BEAM

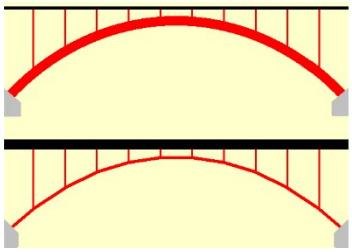
Draw a free body diagram into your jotter and record the results in full sentences.

Bridges

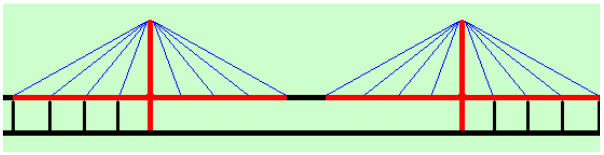
Bridges are a very well known structure, but unlike many other structures, they come in many shapes and forms. Some of the designs which are used are shown below. You can see some using beams, supports and frames.



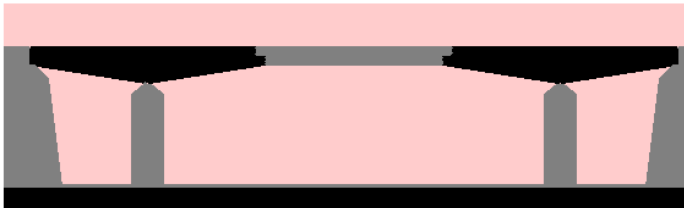
BEAM



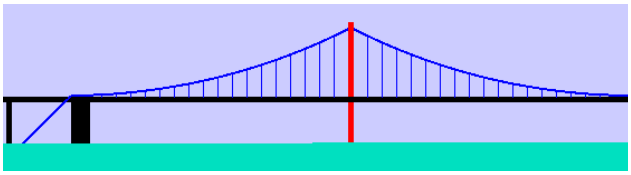
ARCH



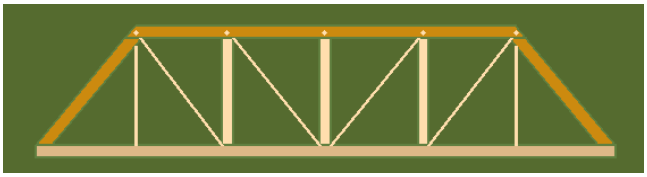
CABLE-STAYED



CANTILEVER



SUSPENSION



TRUSS

14. Problem brief

Using *West Point Bridge Designer*, you are to Design and model a bridge structure which would be suitable to span the 44 m gap and support the two lane road deck a height of 24 m above the river surface.

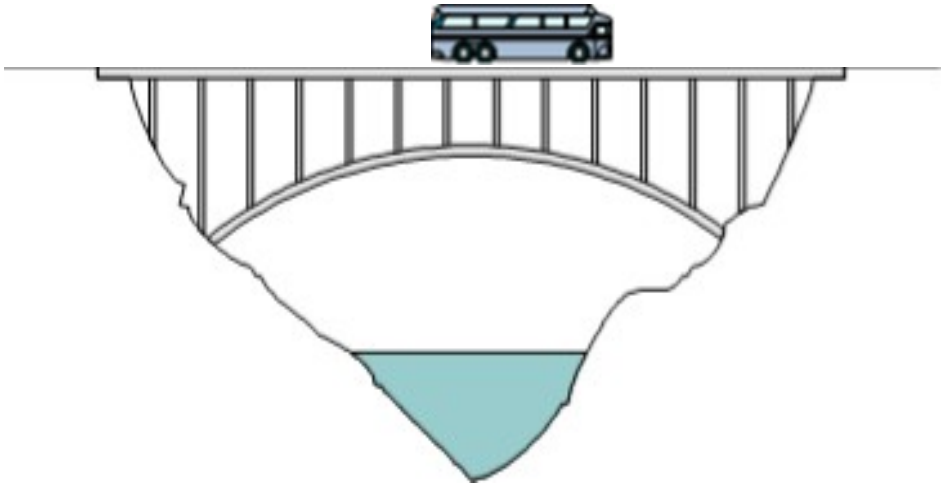
The design of the structure should keep costs to a minimum, but certainly below $200,000.

As part of the report on your design, you are required to show a print of your final bridge design with proof of the final cost, and a print of the test results.

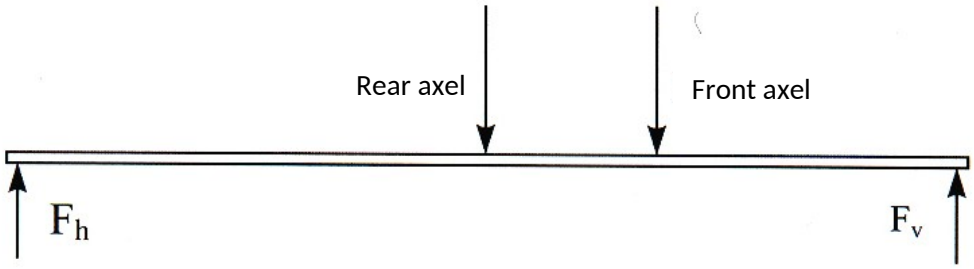
Structures and Free-body diagrams

It is important to isolate different parts of a structure or body form from its adjacent surroundings. In a line diagram this can be done by drawing a free-body diagram, which represents all or part of the structure, showing the forces that affect it.

Example



If all the visual components acting on the structure or body were removed and replaced with their force value. The simplified diagram below would allow a better understanding of how the forces are affecting the structure.



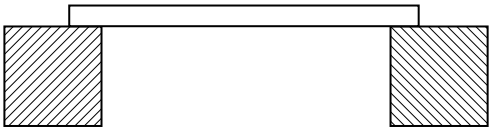
The diagram above of the bus crossing the bridge has been simplified and is called a ***free-body diagram***. The forces representing the bus are shown through where its centre is, and the weight of the bridge is represented through the centre of the bridge. The forces F_h and F_v represent the forces that the supports have on the structure; therefore they also have to be shown. Notice the buses rear axel is located at the mid span of the bridge and the front axel to the right.

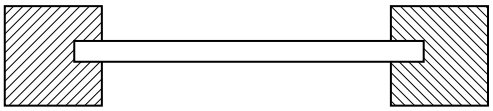
If distances are known, they **MUST** also be shown.

Beams

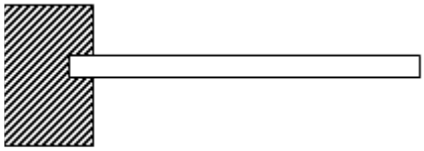
Beams have to be supported differently from lever applications and this determines beam support reactions. Beams, therefore, are supported in a number of ways.

ends.

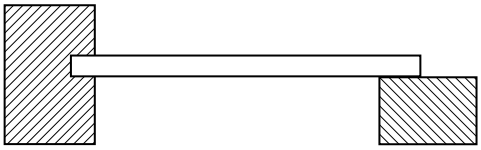




c) **Cantilever**, built in at one end only.



d) Built in at one end and simply supported at the other.



At the point of support, the downward forces acting on the beam are resisted by the forces acting upwards. These upward forces are known as beam reactions, or simply the 'reactions'.

Beam forces (reactions)

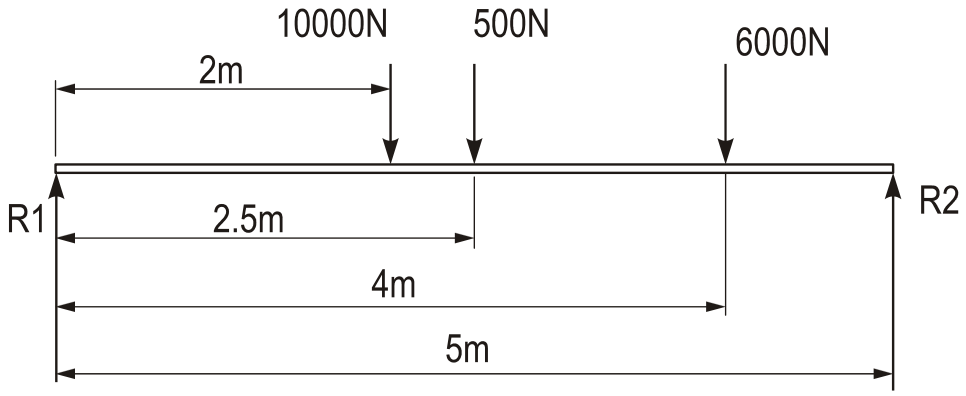
Apart from levers, structural beams and beam related objects are also affected by forces and turning moments. For a horizontal structure to be stable (in equilibrium) when being affected by forces in a vertical plane, the following two conditions must be satisfied.

1. The 'sum' of the downward forces acting upwards must equal the 'sum' of the forces acting downward or left and the right.

2. The 'sum' of the clockwise moments about any point must equal the 'sum' of the anticlockwise moments about the same point.

Worked example

Determine the reactions R_1 and R_2 for the simply supported beam.



Take moments about R_1

$$\Sigma \text{ Clockwise moments} = \Sigma \text{ Anti Clockwise moments}$$

$$(10{,}000\ \text{N} \times 2\ \text{m}) + (500\ \text{N} \times 2.5\ \text{m}) + (6000\ \text{N} \times 4\ \text{m}) = R_2 \times 5\ \text{m}$$

$$R_2 = \frac{20{,}000 + 1250 + 24{,}000}{5}$$

$$\mathbf{R_2 = 9050\ N}$$

Also Σ upwards forces = Σ downwards forces

$$R_1 + 9050 = 10{,}000 + 500 + 6000$$

$$R_1 = 16{,}500 - 9050$$

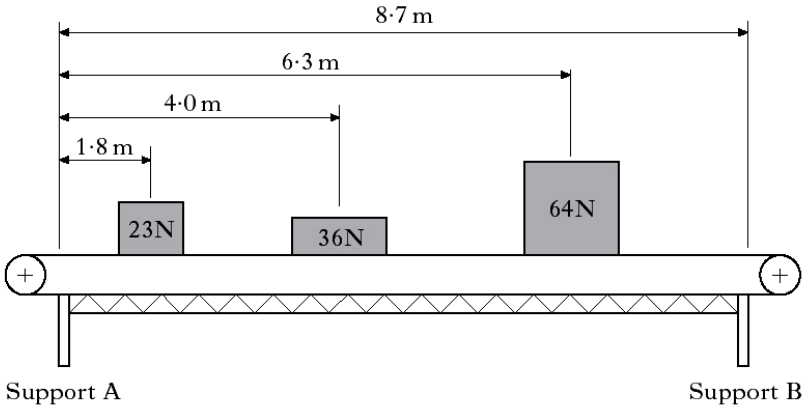
$$\mathbf{R_1 = 7450\ N}$$

Therefore the reactions for the beam supports are

$R_1 = 7450$ N and $R_2 = 9050$ N

15. In your jotters, draw a free body diagram of the pictures shown below, indicating clearly all forces and reactions.

(A) A conveyor belt is used to move packages in a warehouse. When the conveyor belt stopped, packages were in the position shown.



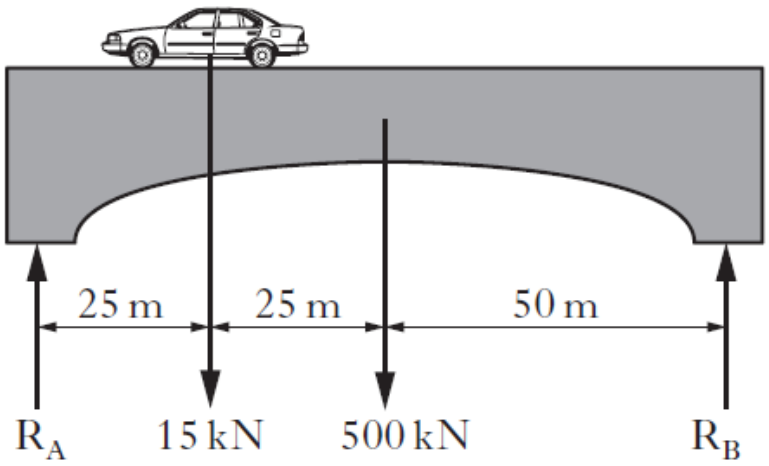
(i) Draw the free body diagram for the system.

(ii) Calculate, by taking moments about support A, the reaction at support B.

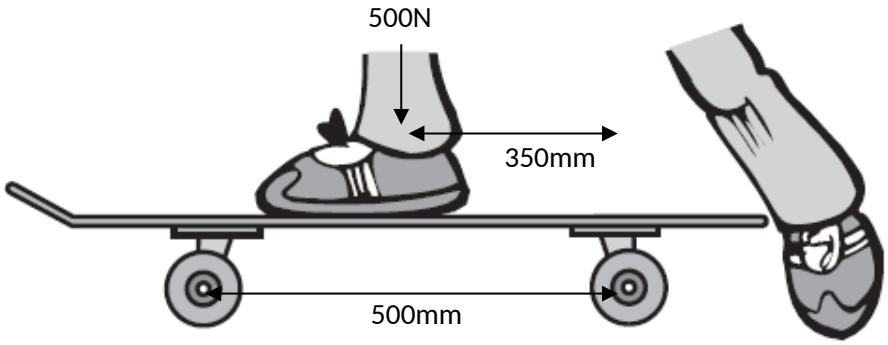
(iii) Determine the reaction at support A.

B) (i) As part of an army training routine, soldiers have to climb across monkey bars 25 metres long. A solider weighing 95kg starts on the left hand side but stops 9 metres along to stop to have a rest. Draw the free body diagram.

C) (i) Draw the free body diagram of the bridge structure shown below.

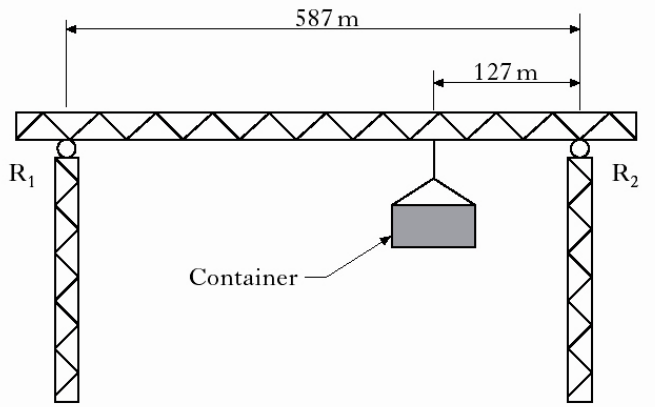


16. Produce a free body diagram for the following questions in your jotter and calculate.



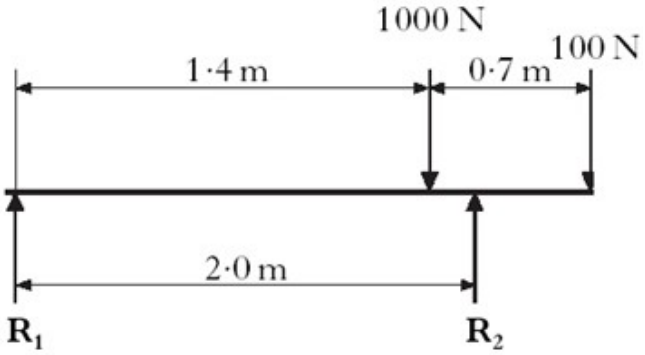
(i) Draw the free body diagram.
(ii) Calculate, by taking moments about R_2, the size of the reaction R_1.
(ii) Determine the size of the reaction R_2.

17. **Exam question.** The structure below is used to transport a 500kg container.

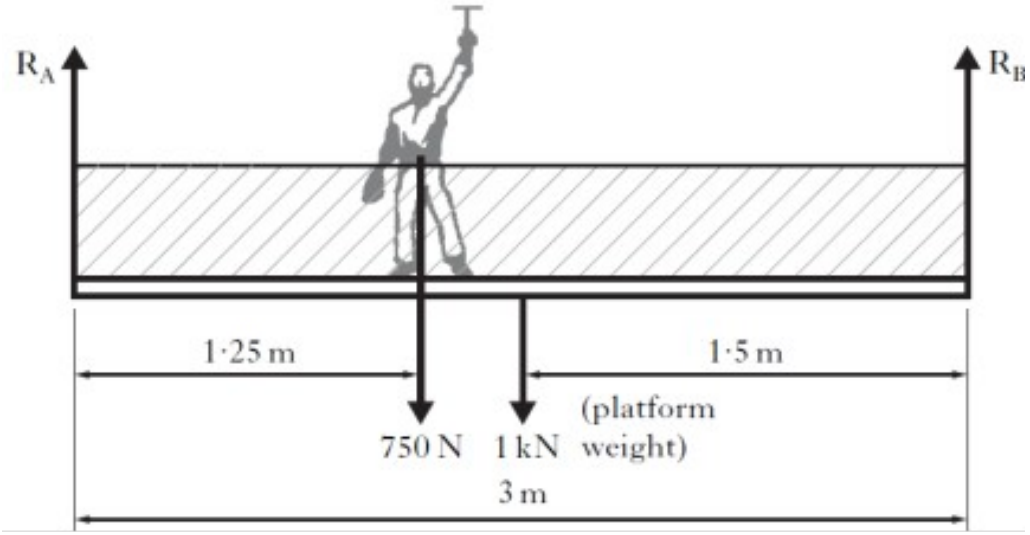


(i) Calculate the weight of the 500 kg container.
(ii) Draw the free body diagram of the structure, showing all forces and distances.
(iii) Calculate, by taking the moments about R_2, the size of the reaction R_1
(iv) Determine the size of the reaction R_2.

18. **Exam question.** A free body diagram for a car trailer is shown below. Taking moments about R_1, calculate the reaction R_2. **(show all working and units)**

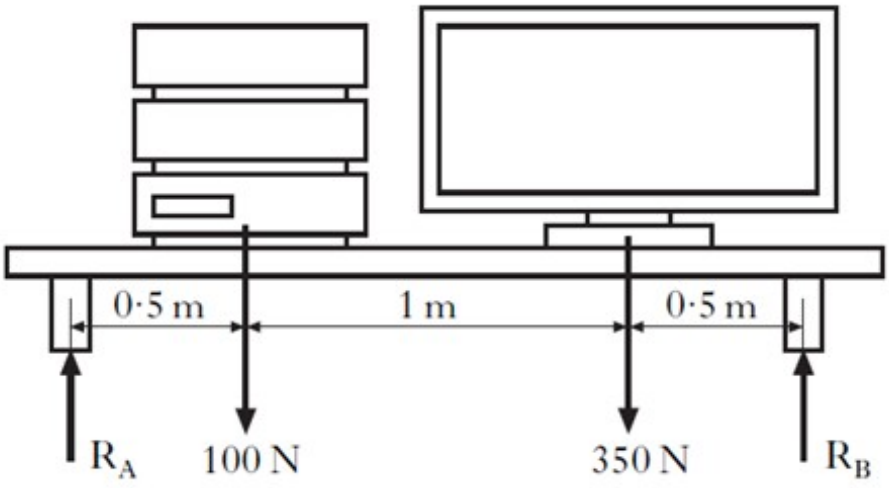


19. The image below shows a window cleaner's platform.



i) Draw a free body diagram of the system.
ii) Calculate the reaction force R_A (take moments about R_B)
iii) Calculate the reaction force R_B.

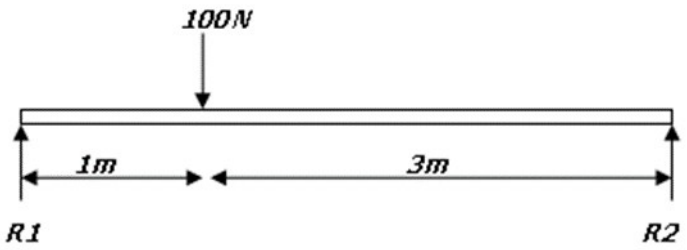
20. A unit used to support the DVD player, an entertainment system and television is shown.



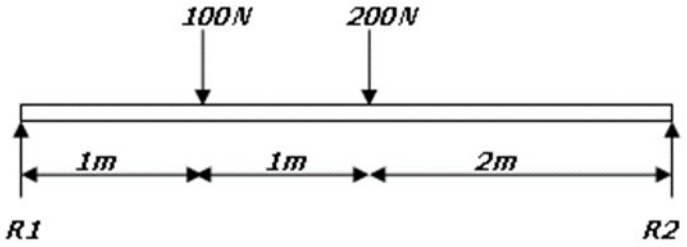
i) Draw a free body
ii) Calculate the reaction force R_A (take moments about R_B)
iii) Calculate the reaction force R_B.

21. Copy the free body diagrams into your jotter and determine the reactions R_1 and R_2 for the simply supported beams in the following examples (beam weights will be ignored in this case).

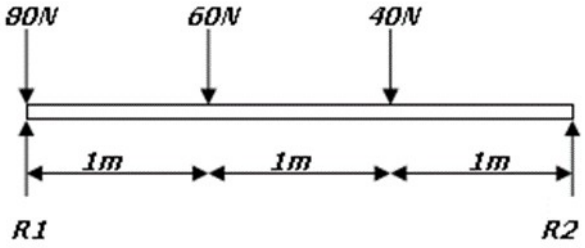
A)



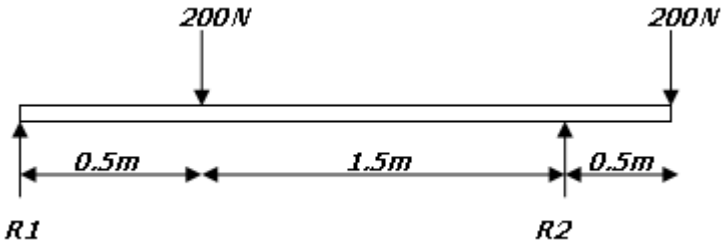
B)



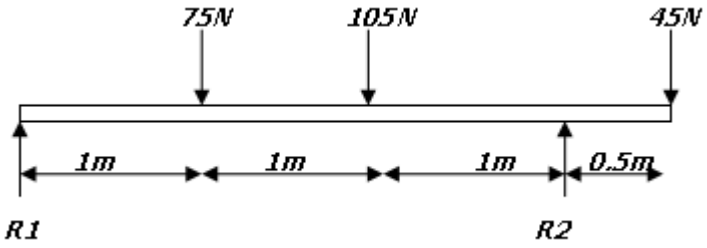
C)



D)

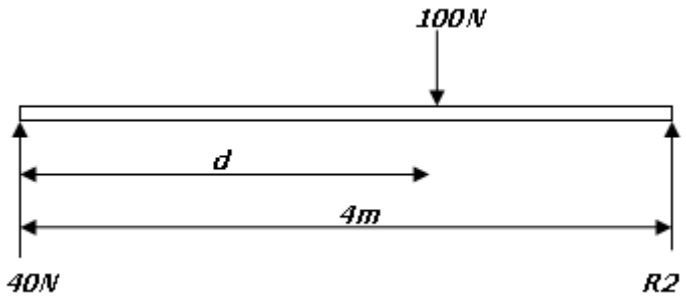


E)

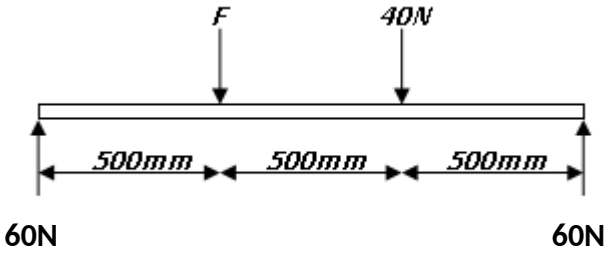


22. Copy the free body diagrams into your jotter and calculate the missing quantity. Use the principle of moments and assume that each beam is in a state of equilibrium.

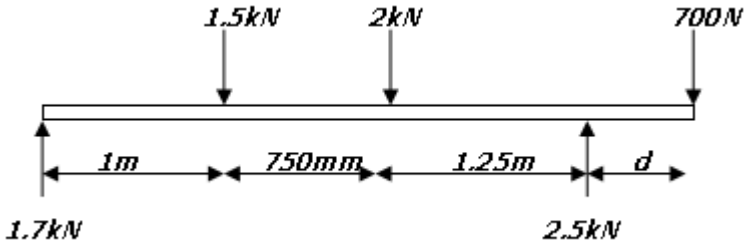
A)



B)



C)



D)

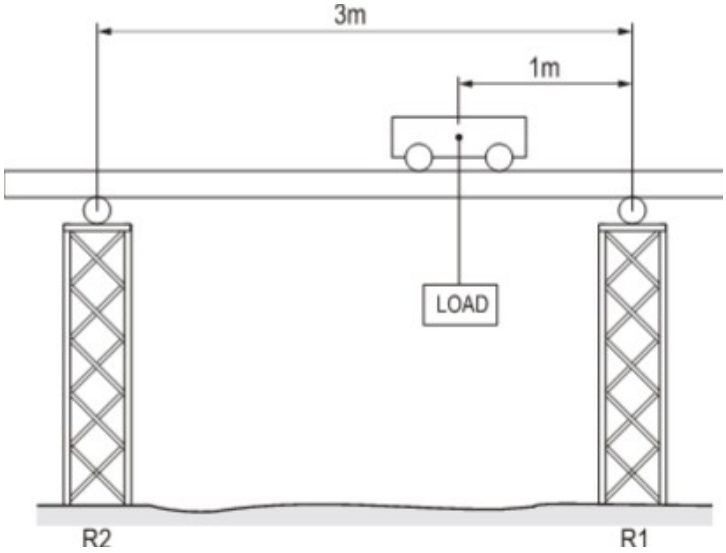
E)

23. A beam is simply supported at each end with a span of three metres. The beam carries a small lifting device having a weight of 1 KN. Answer in jotter.

a) Complete a suitable free-body diagram.

b) When the lifting device is positioned at the ***mid-point*** of the beam and carries a casting weighing 2.5 kN, what are the reactions at R_1 and R_2?

c) When the lifting device is positioned one metre from one end and carries a machine component weighing 6 kN, what are the reactions at R_1 and R_2?

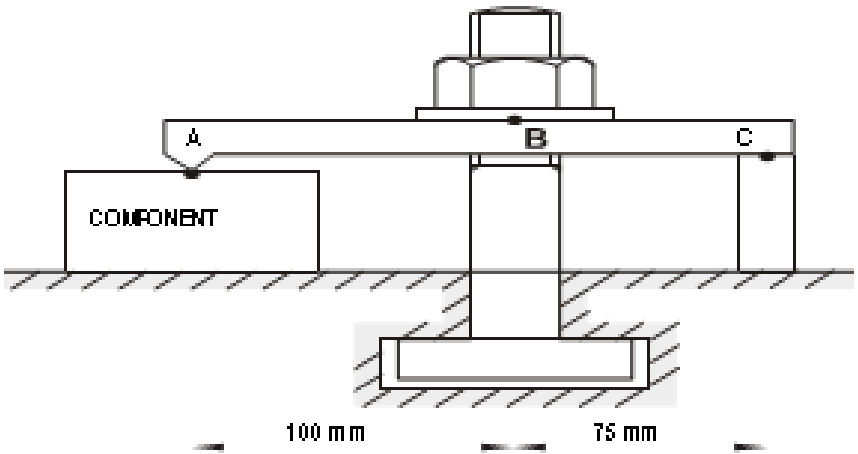


24. The figure below shows a clamp on a milling machine table for fixing a component for machining. A clamping force of 1200 N is applied by the bolt to the component and rear-distance piece when the nut is fully tightened. Answer in jotter.

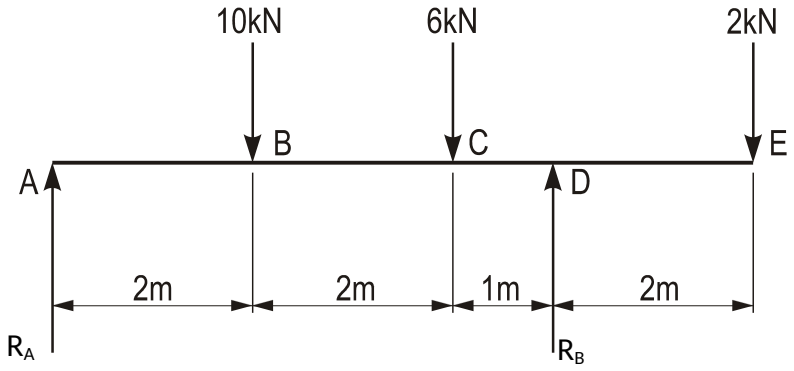
a) Draw a free-body diagram to show the arrangement of the forces.

b) Calculate the clamping forces on the component (R_A) and the distance piece (R_C).

c) How could the arrangement be altered to give a bigger clamping force on the component?



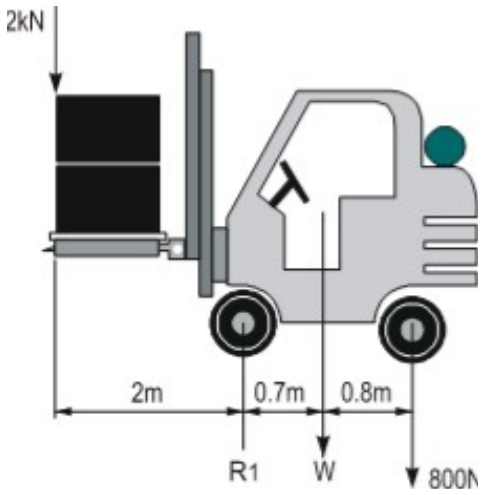
25. The diagram below shows a free-body diagram of a horizontal beam, seven metres long, which is part of a bridge structure. The beam is simply supported at A and B.



a) Calculate, by taking moments (fulcrum) about R_B, the size of the reaction R_A.

b) Determine the size of the reaction R_B.

26. The forklift truck must have a minimum downward force of 800 N acting through the rear wheels. Answer in jotter.

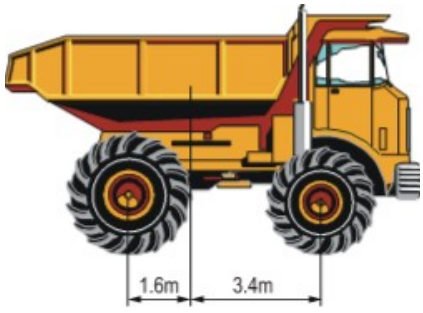


a) Draw an appropriate free-body diagram.

b) Calculate the weight required to balance the load on the lift with $R_2 = 0$ N.

c) Find the additional weight acting through the centre of gravity of the truck to produce 800 N at the rear wheels.

27. The total downward load when the truck shown below is empty is 30kN and when fully loaded 55KN. Draw a suitable free-body diagram. Find the reaction in each of the axles when the truck is empty and when fully loaded. Answer in jotter.

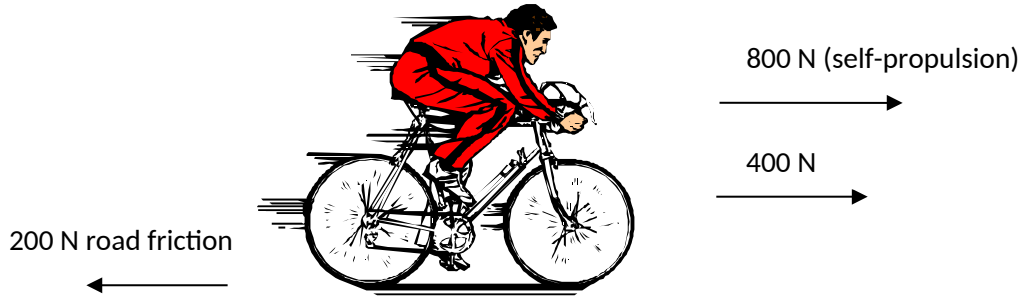


Vectors

Force is a vector quantity and has both magnitude and direction. This means it is often convenient to represent a force by a line, that is, a vector quantity, which is sometimes easier to understand visually. The direction of the force may be indicated by an arrow-headed line, with the length of the line drawn to scale to represent the size of the force. This line is called a **vector**.

Worked example

The cyclist pedalling with a force of 800 N is being assisted by a tail-wind of 400 N, but the friction from the road surface measures 200 N.



The overall effect will be: 800 N + 400 N – 200 N = 1000 N (or 1 kN).

A suitable scale would be selected – possibly 10 mm to represent 100 N – and using this scale each force is drawn in turn, one following on from the other. This is called a **'vector diagram'**

800 N + 400 N – 200 N = 1000 N or 1 kN

This arrow which points to the right is 100mm in length.

A vector diagram

When the three forces are added together (above), they can be replaced by a single force that has the same effect, called the '**resultant**'.

RESULTANT = 1 kN

28. Draw the resultant vector diagrams for each picture, using a suitable scale. Answers in jotter.

A)



B)



C)



Vectors at angles

Vectors are also used to find the resultant of two forces that are inclined at an angle to each other.

In the example below, the resultant of the two forces can be found by drawing two vectors. First choose a suitable scale and draw the two vectors CA and CB.

Scale: 10 mm = 10 N
28 mm = 28 N = CA
60 mm = 60 N = CB

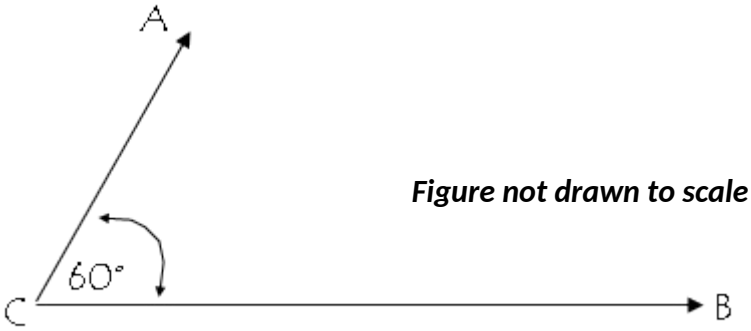


Figure not drawn to scale

The bigger the scale, the more accurate the vectors.

From A, draw a line parallel to CB, and from B draw a line parallel to CA. Call the point where the two lines intersect point D. Now draw a line from C to D. A line drawn from C to D is the resultant of the two forces CA and CB. Try it yourself on grid paper.

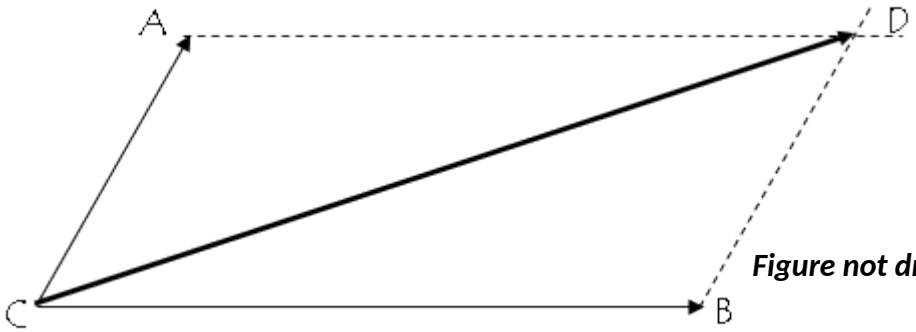


Figure not drawn to scale

The resultant has a magnitude of 77N by measurement.

Equilibrium

As you already know certain conditions must apply with structures in order to create stability. The resultant is made up of the combined forces that are trying to move an object/structure in a set direction. If such a force were applied without an opposing force then major problems would occur. Structures have to remain in a stable or balanced state known as equilibrium. There are 3 types of balancing that must exist if an object is to remain in equilibrium: **(ΣM = 0)**

Horizontal forces **Vertical forces** **Rotational forces**

The general conditions of equilibrium are as follows:

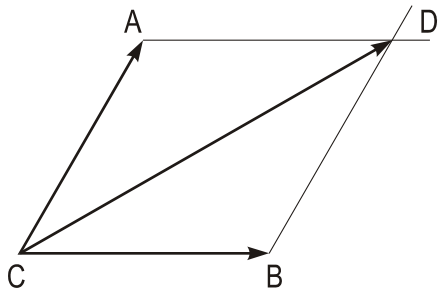
Upward forces = downward forces

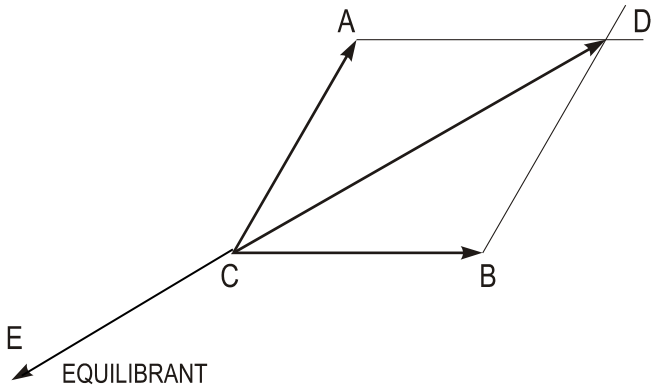
Leftward forces = rightward forces

Clockwise moments = anticlockwise moments

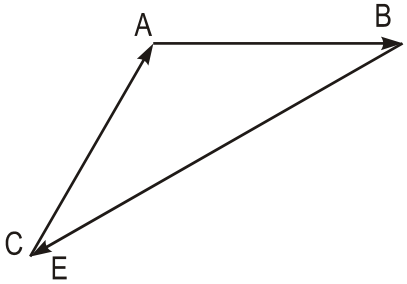
Worked example 1

Consider the two forces acting below. Are they in equilibrium? It is easily seen that a force must be added acting downwards to the left, but we cannot tell from this exactly how large this force must be or its exact direction.



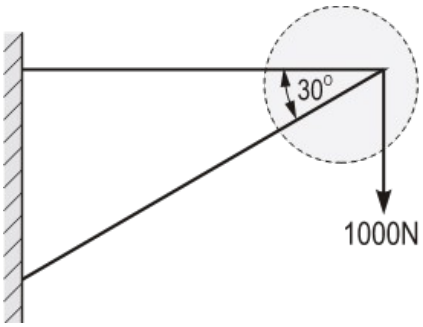


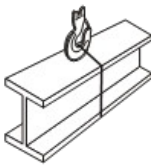
The '**resultant**' has been drawn and it can be seen that to balance it, the equilibrant CE is required. If the forces F_1 and F_2 are drawn as in the image above, then it is much easier to obtain the 'equilibrant' by completing the simplified triangle as shown below.

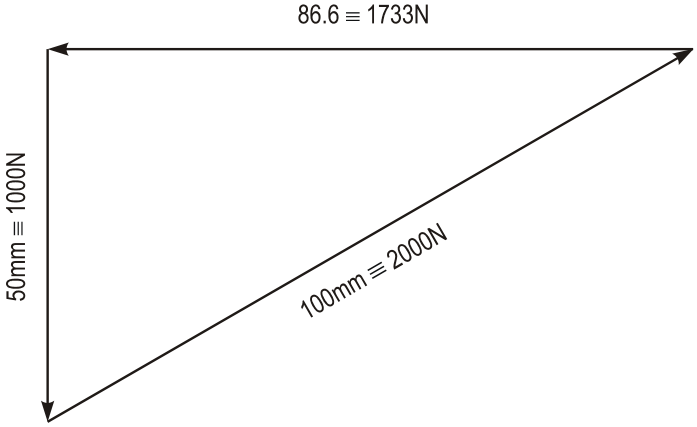


Worked example 2

A crane is fixed against a wall, as shown below. Find the forces in the compression and tension members.







To find the forces created in the tension and compression members by the 1000N load, draw the triangle from the area circled. Select a suitable scale and then draw the known force first, the 1000N load. A line is drawn through one end of the load line parallel to one of the unknown forces. Another line is drawn through the other end, parallel to the second unknown force. By measuring each line, the size of each force can be found. (Note: the arrowheads must follow round the triangle.)

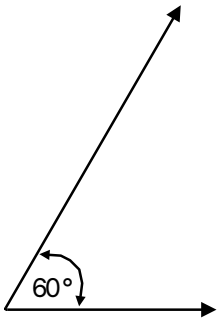
Scale: 10 mm = 200 N

The compression member = 2000 N

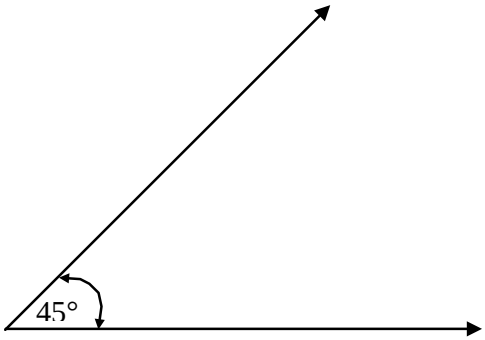
The tension member = 1733 N

29. Using the diagrams below, calculate the resultant force for each if 10mm = 10N.

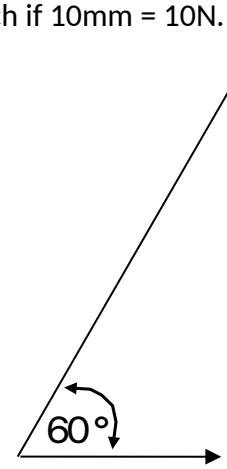
A)



B)



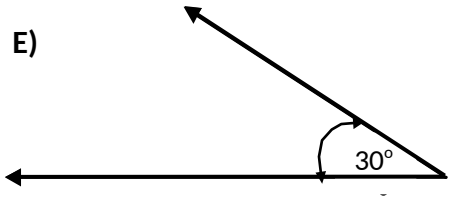
C)



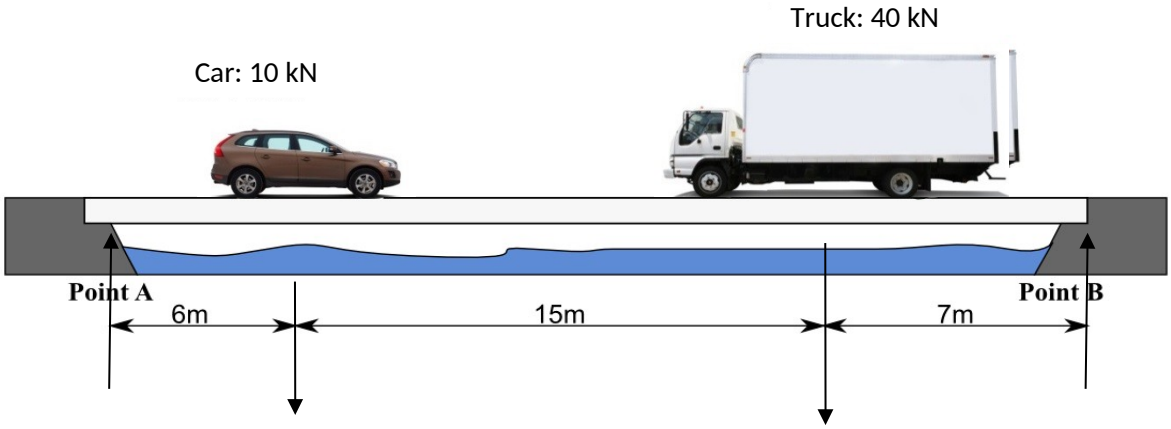
D)



E)

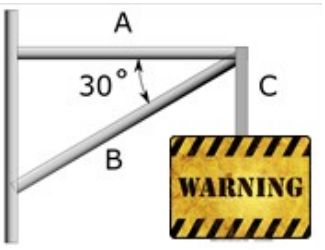


30. A simplified diagram of traffic on a bridge is shown below. The car axels have a space of 4.2 metres and the Truck's axel space is 7.5 metres. Assume the vehicle loading is split equally across the axels.

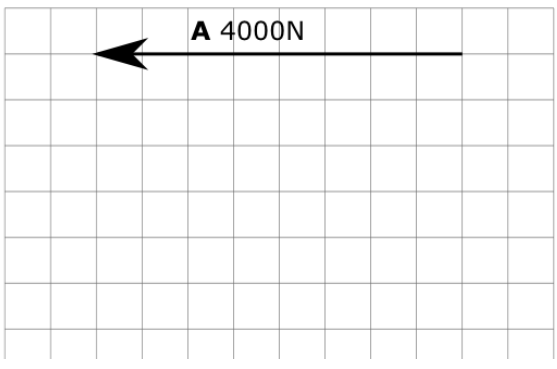


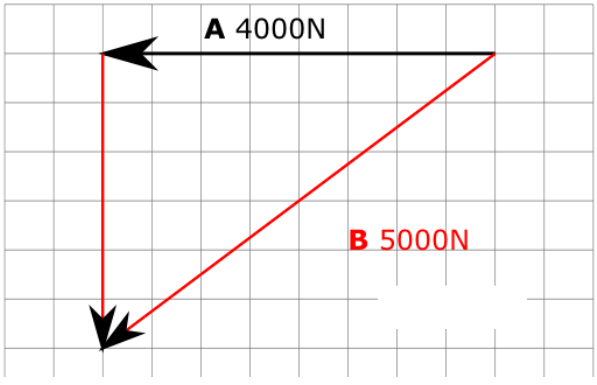
a) Draw a free body diagram of the bridge with the vehicle loading. Answer in jotter.

b) When repairs are being conducted on the bridge, a sign is used to warn motorists and is supported by a frame as shown.

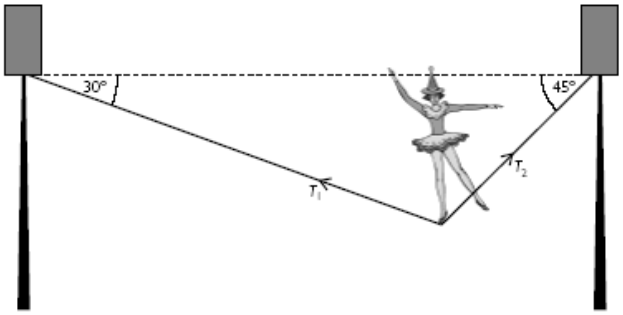


The force in part A is 4000 N.
The force in part C is 3000 N.
Every square = 500N
Determine the size of the force in part B by completing the triangle of forces diagram below:





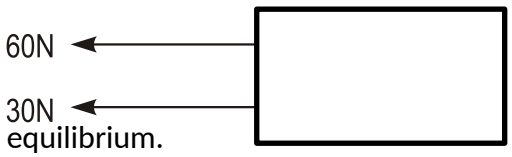
31. A circus high wire act is shown. The high wire performer has a weight of 750N and is at rest. Determine graphically the tension (T_1 and T_2) on the wire.



Equilibrium tasks

1. Study the following statements and cross out the incorrect answer.

- A body that is accelerating is in a state of equilibrium. TRUE/FALSE
- For a body to be in a state of equilibrium it is necessary only for the vector sum of the forces acting on it to be zero. TRUE/FALSE
- A resultant force is a single force that can replace two or more forces. TRUE/FALSE
- If two or more forces are replaced by a resultant force, the effect on the body is changed. TRUE/FALSE
- An equilibrant force is the force that, if applied to a body, will cause the body to be in a state of equilibrium. TRUE/FALSE
- The equilibrant force is identical to the resultant force. TRUE/FALSE



2. Two forces are acting on a body as shown.

a) Explain two conditions necessary for a structure or body to be in equilibrium.

b) Graphically redraw to indicate their size and direction at a scale of 1N = 1mm

c) Graphically indicate the resultant of the two forces at a scale of 1N = 1mm

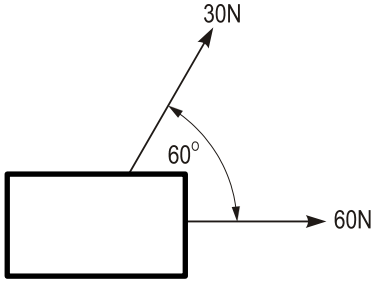


3. Two forces are acting on a body as shown.

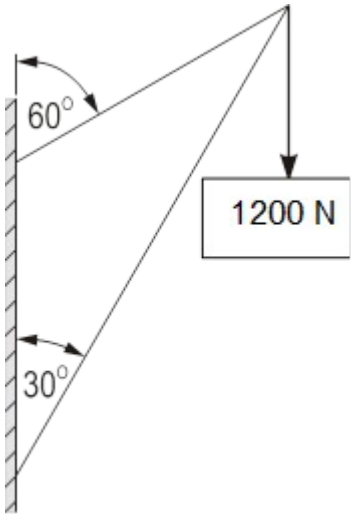
a) Graphically redraw to indicate their size and direction at a scale of 1N = 1mm.

b) Calculate the resultant and direction of the two forces.

4. What are the resultant and equilibrant of the two forces affecting the system opposite?



5. A small crane is used on a fishing trawler to lift cases of fish to the dock. The weight of the lift is 1200 N. Determine the size and direction of the forces in each of the crane members. (Use a scale where 10 mm represents 200 N.)



6. A weight of 2000 N is suspended by a rope attached to a hook firmly fixed to a roof joist. A second rope is attached to the vertical rope and pulled horizontally until the rope makes an angle of 30° to the vertical as shown. Determine the horizontal pull on the rope and the force on the hook.

